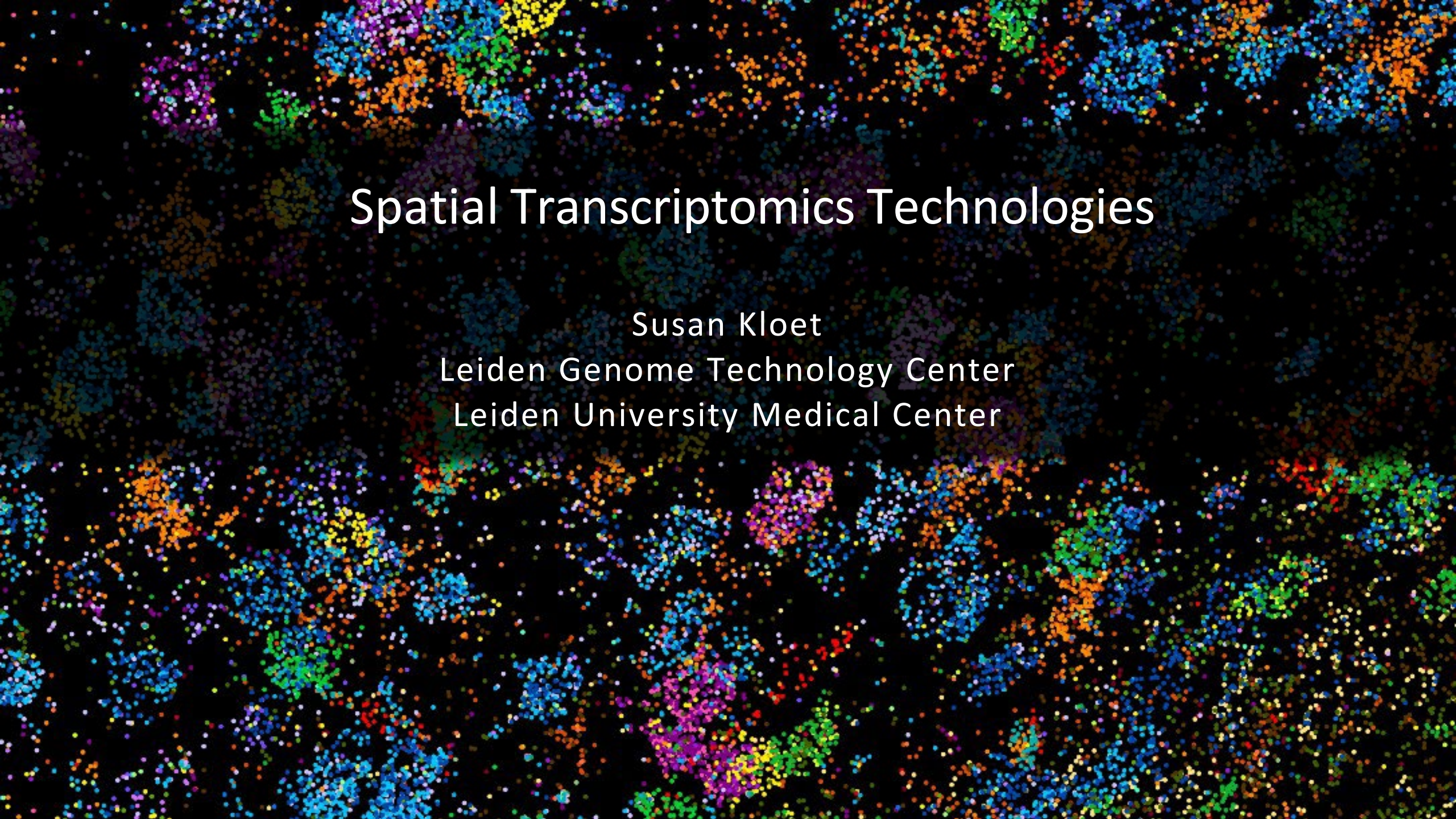




Spatial Transcriptomics Technologies

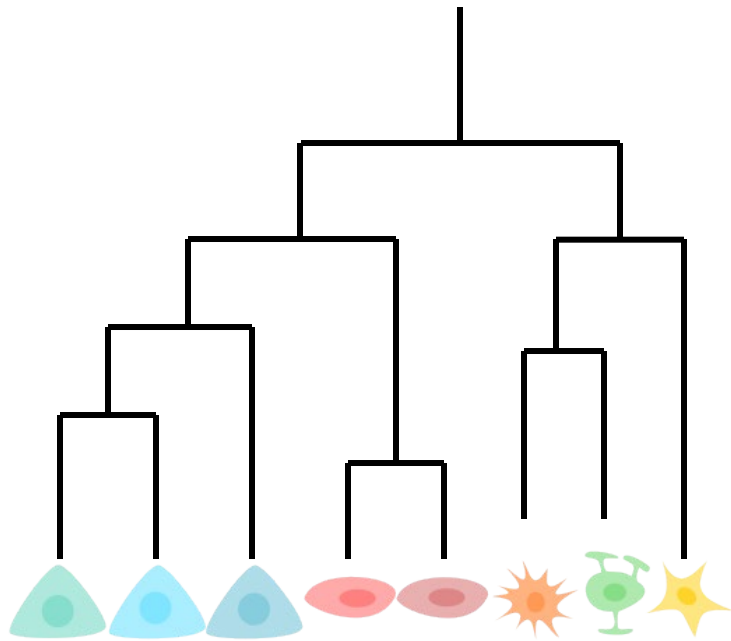
Susan Kloet

Leiden Genome Technology Center

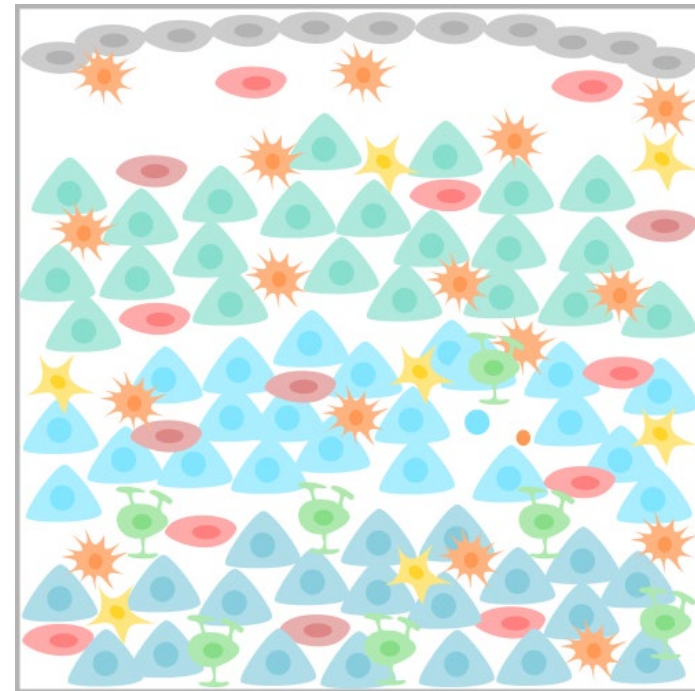
Leiden University Medical Center

Single cell RNA sequencing

Cellular complexity

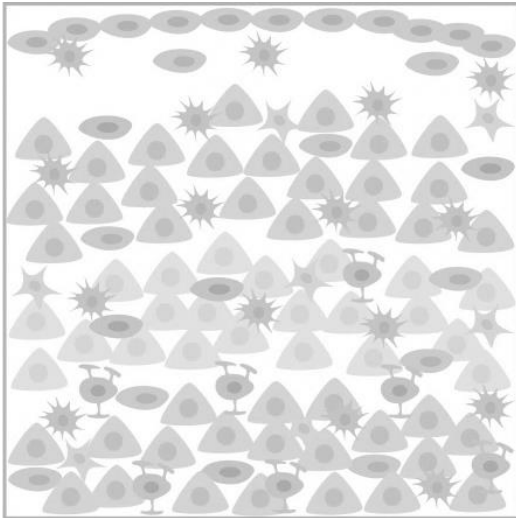


Tissue architecture

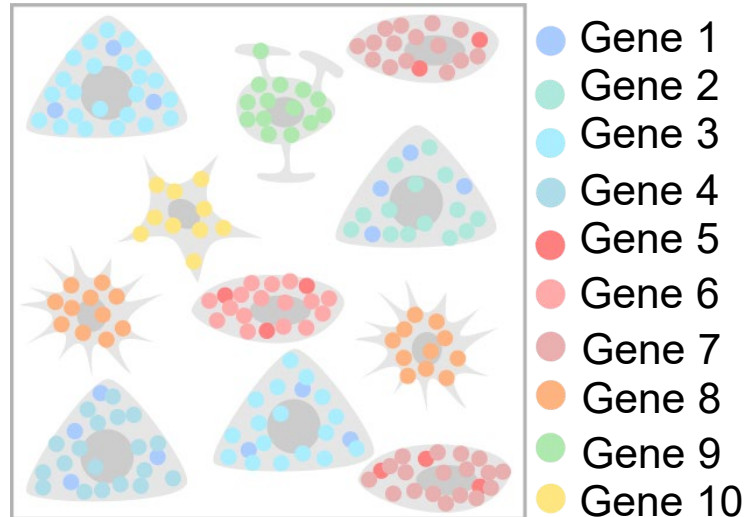


Spatial RNA detection

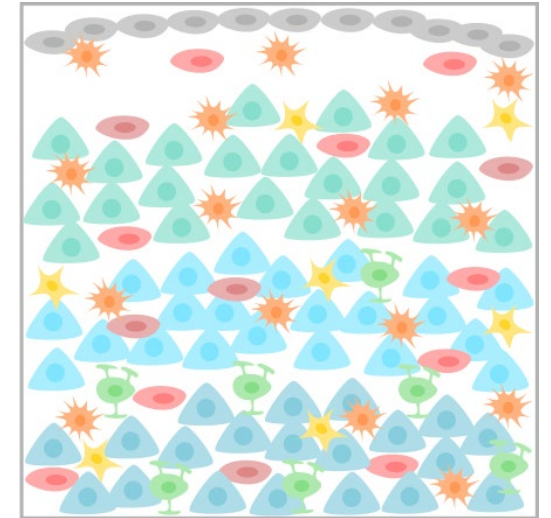
Complex tissue



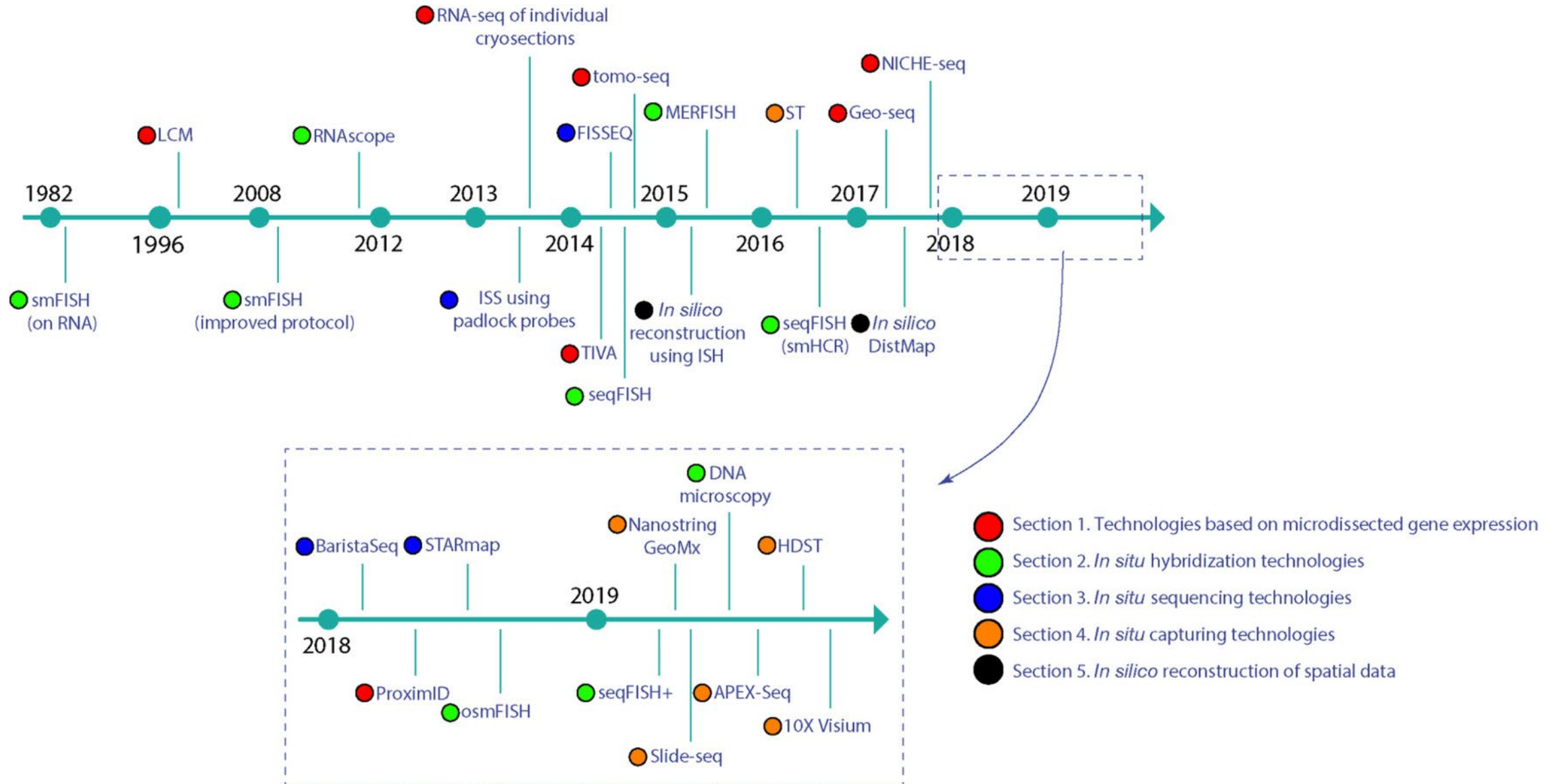
Spatial measurement



Spatial cellular atlas



History of spatial approaches



Technology development continues...



We will cover

- Description of spatial platforms
- Cell segmentation and segmentation-free approaches
- Pre-processing, QC, normalization
- Data integration
- Domain identification and niche reconstruction
- Spatial annotation
- Cell-cell communication

Participation time!

Who has (or will soon generate) spatial transcriptomics data?

Who has (or will soon generate) spatial metabolomics data?

Who has (or will soon generate) spatial proteomics data?

Who has (or will soon generate) multimodal spatial data?

Something else?

Goals for today's lecture

Categorize spatial transcriptomics technologies

Understand how these technologies work

List the strengths and weaknesses of these technologies

Spatial RNA detection



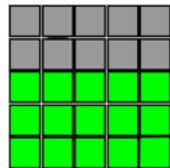
Resolution



Sensitivity



Number of targets

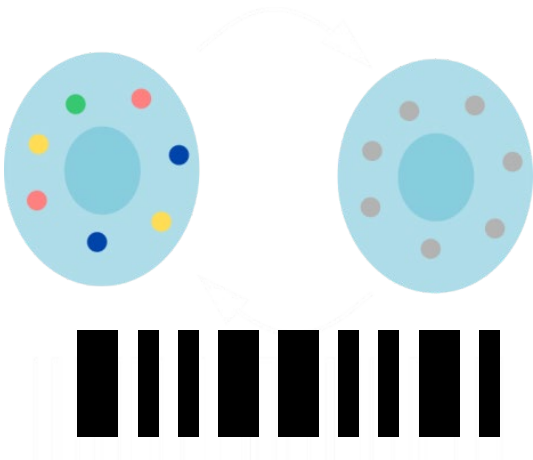


Area

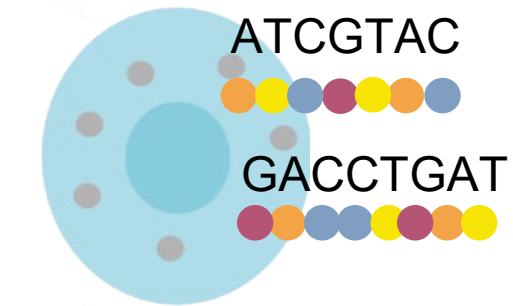
Spatial RNA detection

Microscopy

Barcoded FISH

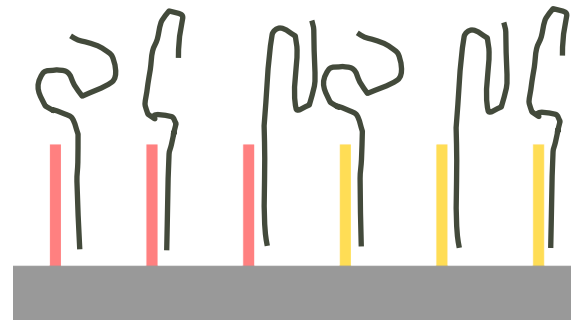


in situ sequencing

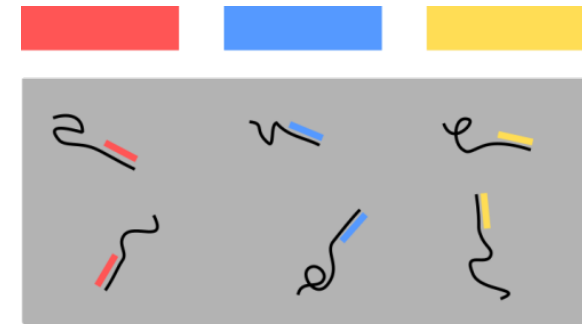


Sequencing

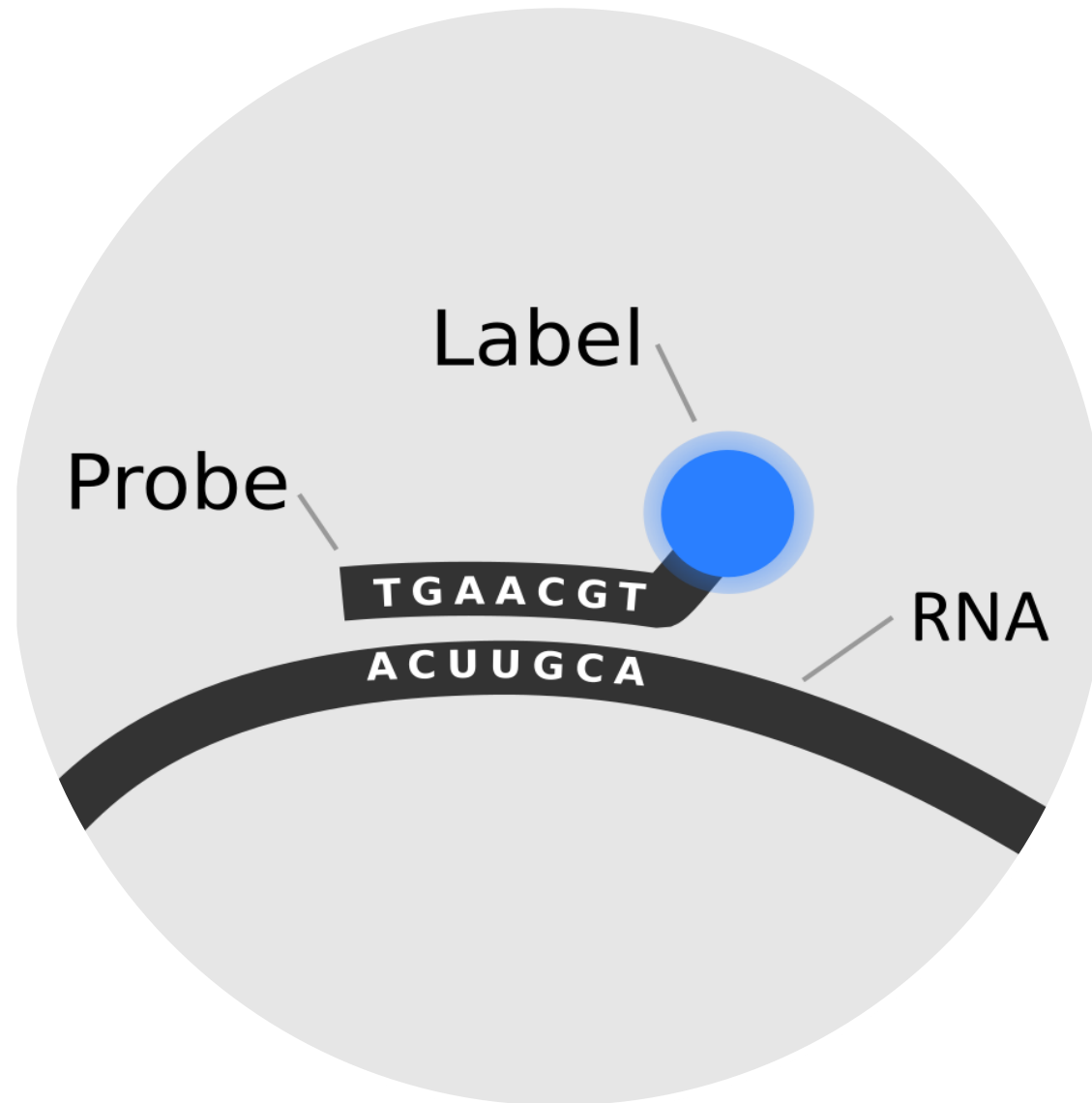
Spatial sequencing



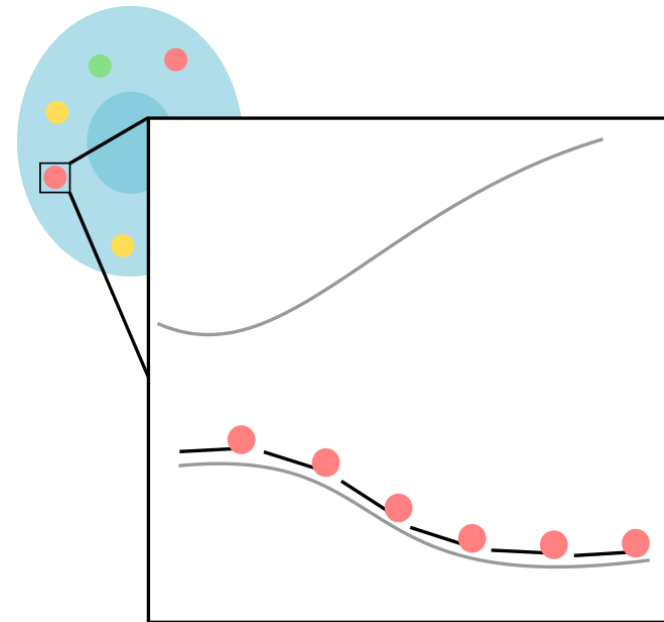
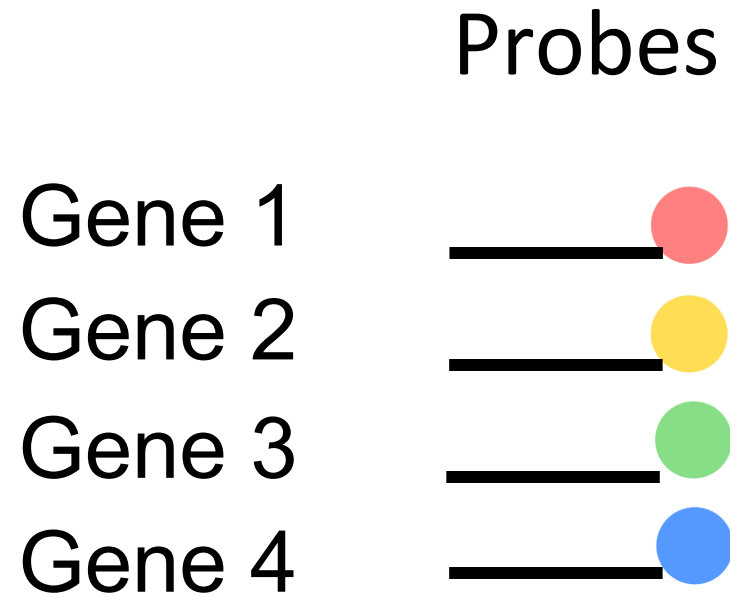
Spatial tagging



Fluorescent *in situ* Hybridization (FISH)



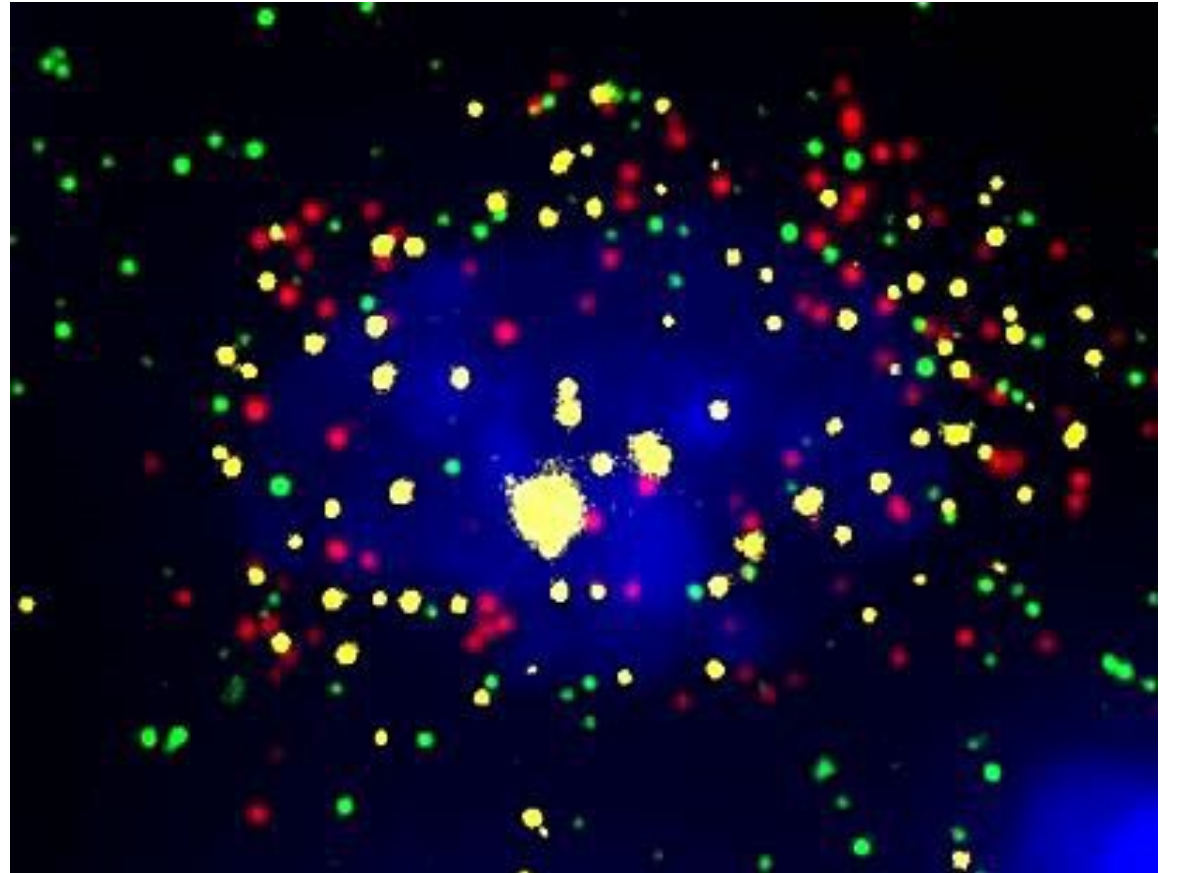
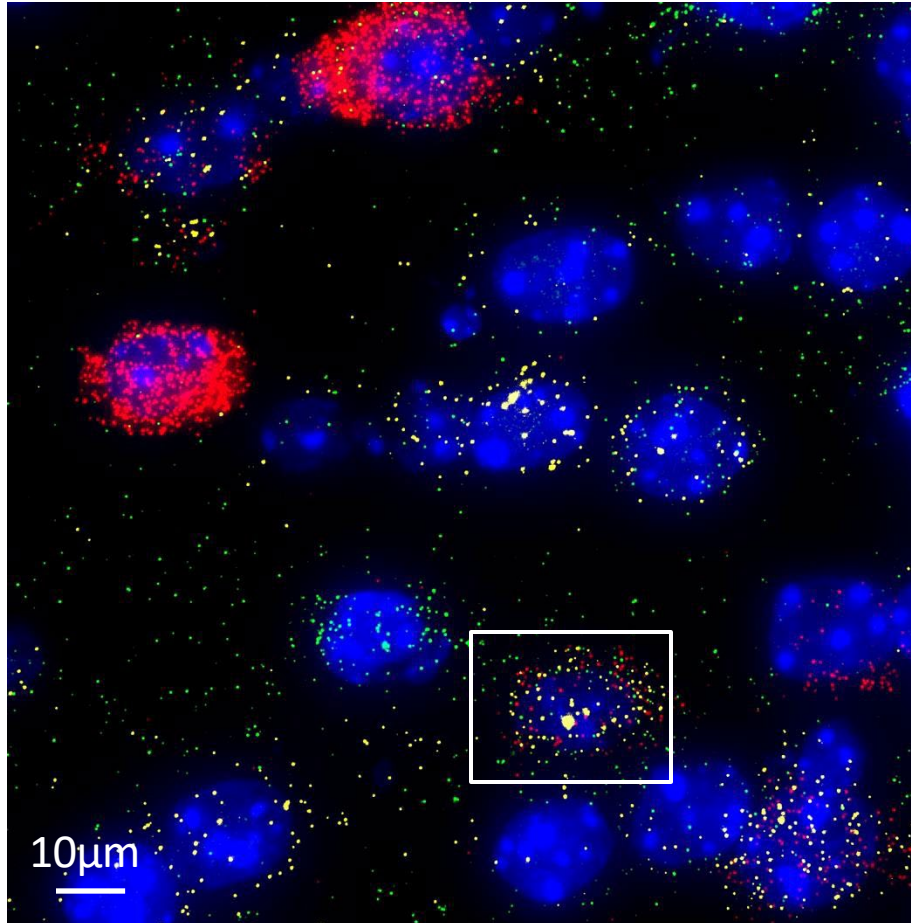
single molecule FISH (smFISH)



Femino *et al.* Science 1998
Raj *et al.* Nature Methods 2008

single molecule FISH (smFISH)

Gene 1 Gene 2 Gene 3 DNA



Breaking the color barrier

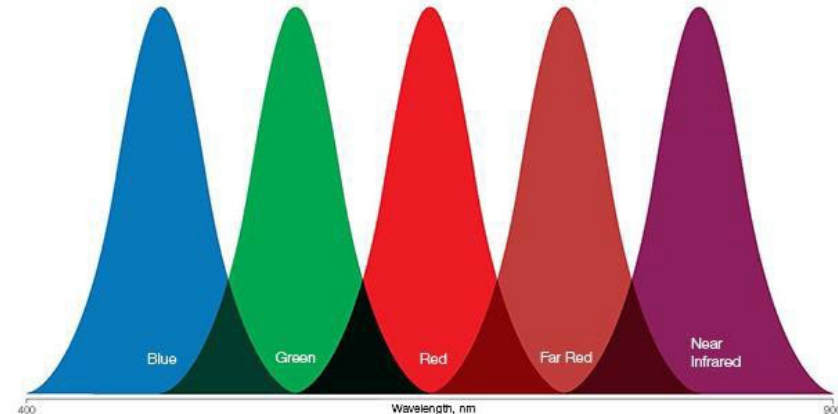
~20.000 genes

4-7 colors

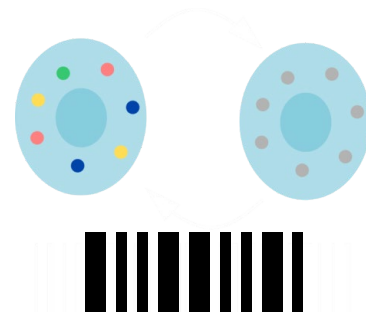
Solution:

- Repeated staining on same sample
- Barcoding

Limited fluorophores



Reprobing same molecule



Round 1



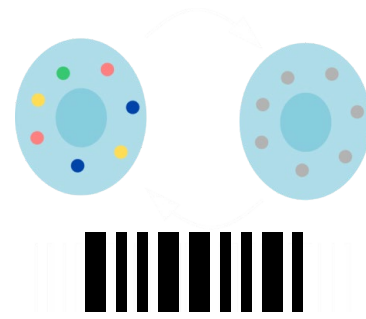
Stripping



Round 2

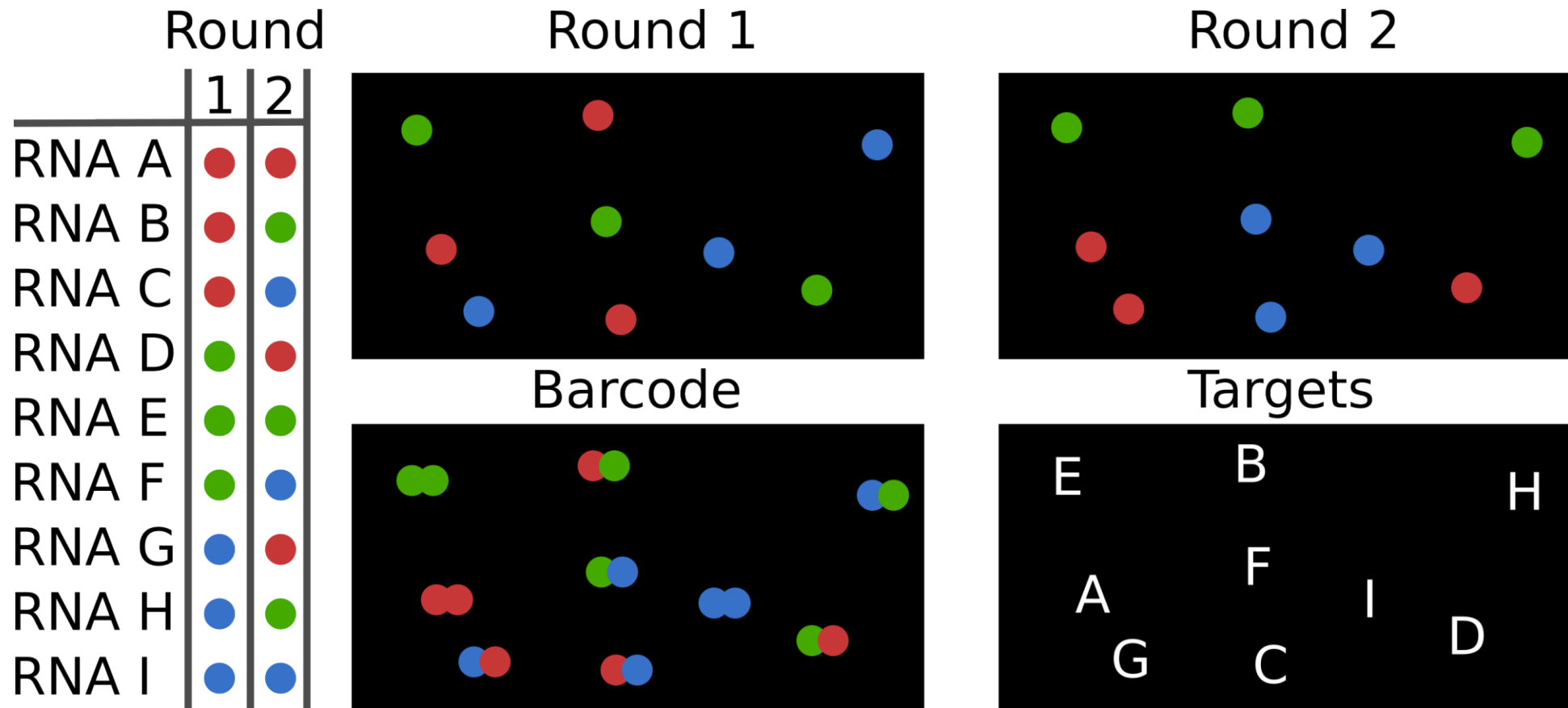
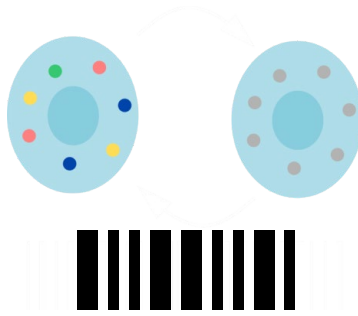


Barcoding



	Round	
	1	2
RNA A	●	●
RNA B	●	●
RNA C	●	●
RNA D	●	●
RNA E	●	●
RNA F	●	●
RNA G	●	●
RNA H	●	●
RNA I	●	●

Barcoding



Barcoding

Scaling-up: $targets = f^n$

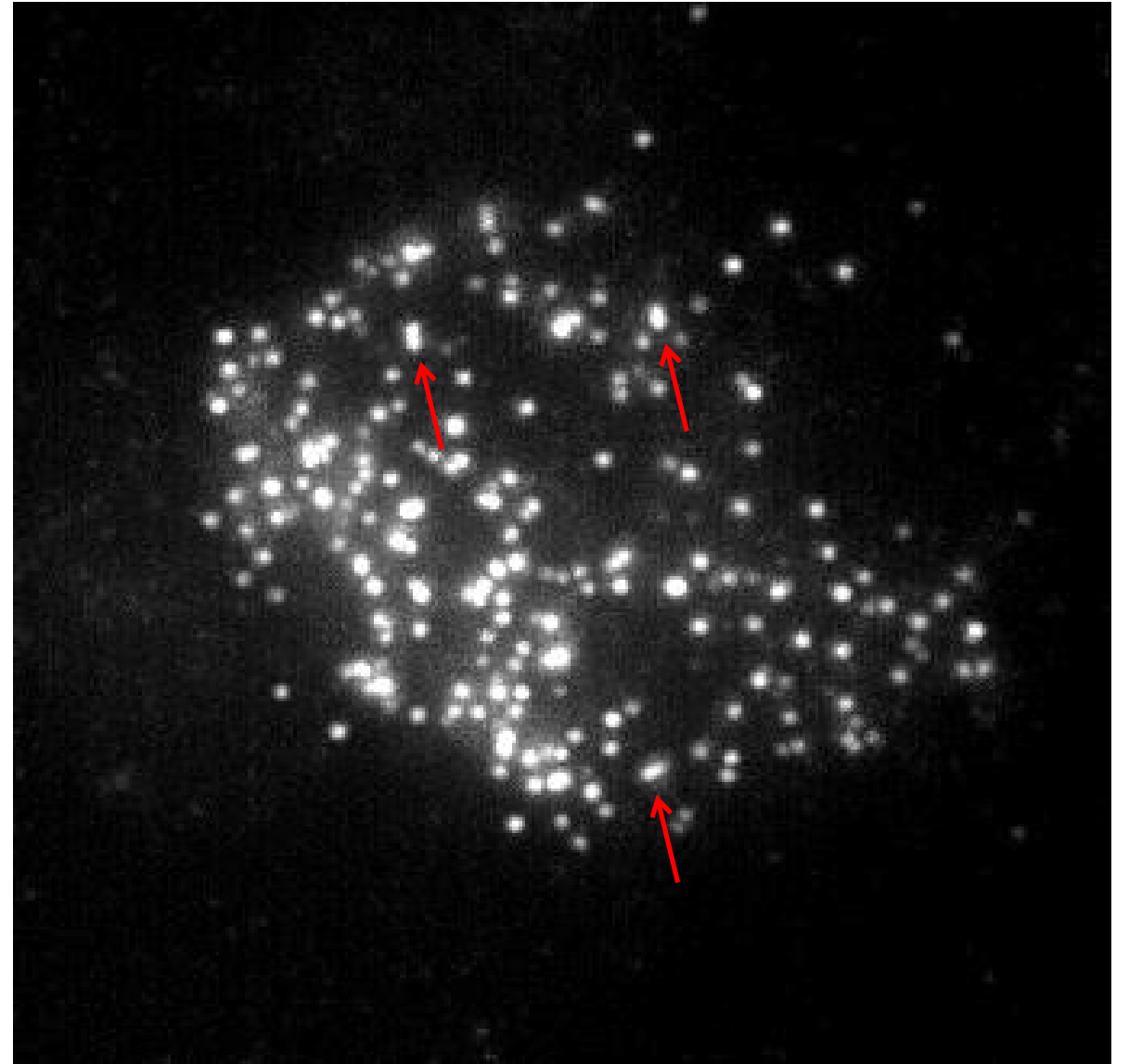
$$4^8 = 65,536$$

Problems:

- Optical density
- Errors

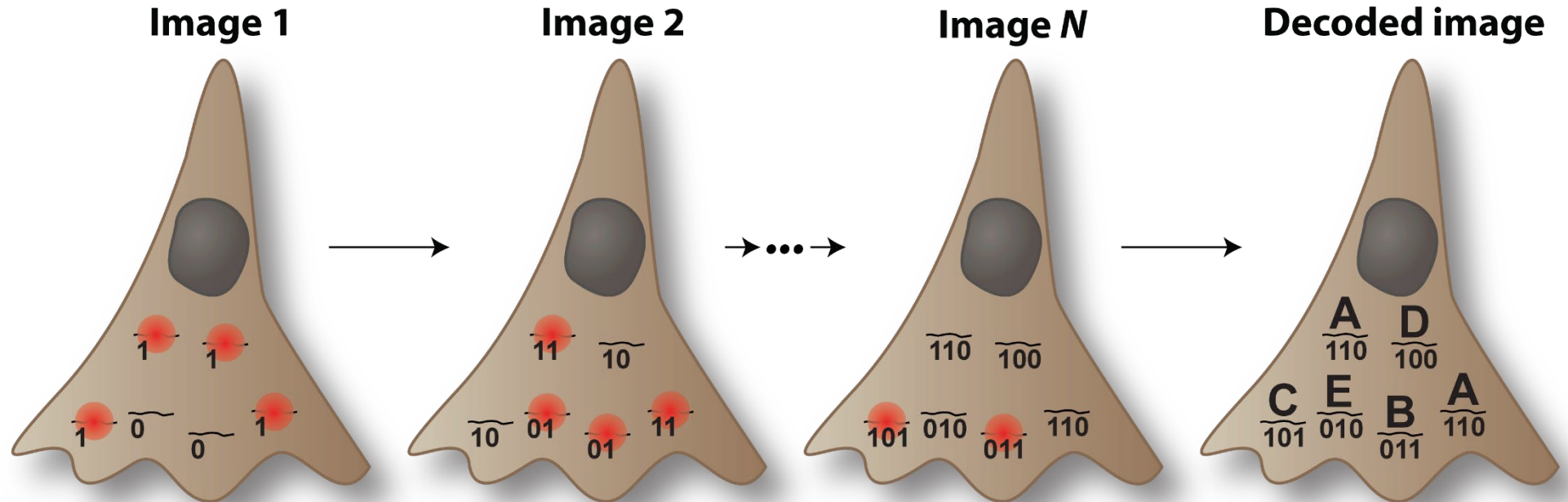
Solution:

- Sparse barcodes

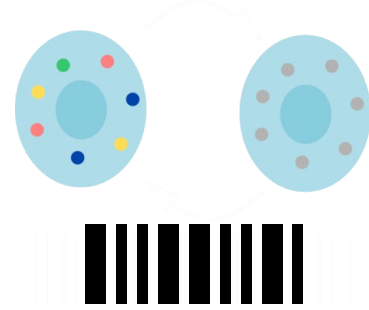


Sparse barcoding

RNA	Barcode
<i>A</i>	11...0
<i>B</i>	01...1
<i>C</i>	10...1
<i>D</i>	10...0
<i>E</i>	01...0
<i>F</i>	00...1
<i>G</i>	11...1



Barcoded smFISH



Methods: MERFISH, seqFISH, EEL FISH

Companies: Vizgen, Spatial Genomics, Nanostring/Bruker, Resolve



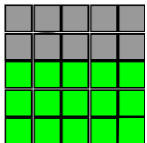
Resolution: Diffraction limited (150-300nm)



Detection efficiency: 70-90%



Gene throughput: 100 - 19,000

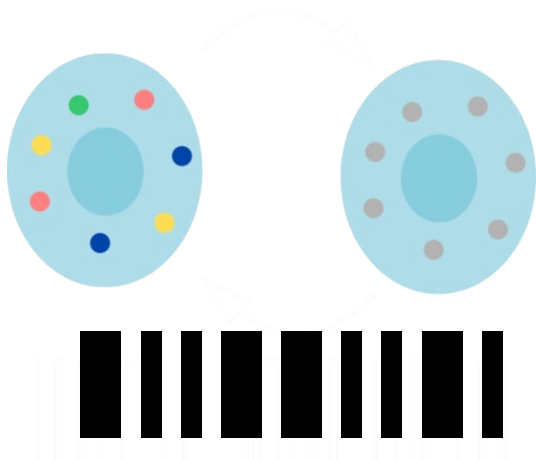


Spatial throughput: several mm² - cm²

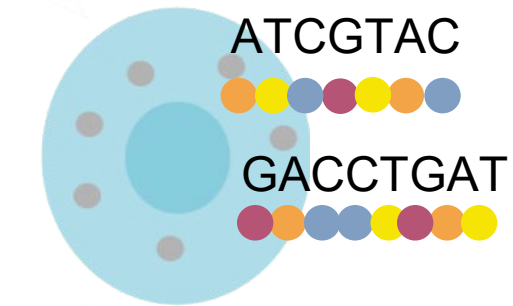
Spatial RNA Detection

Microscopy

Barcoded FISH

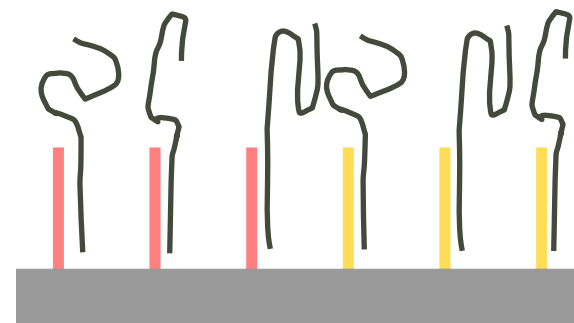


in situ sequencing

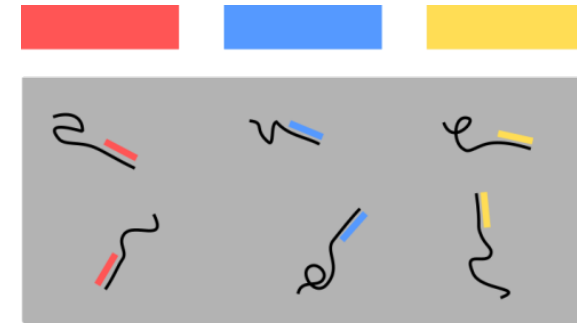


Sequencing

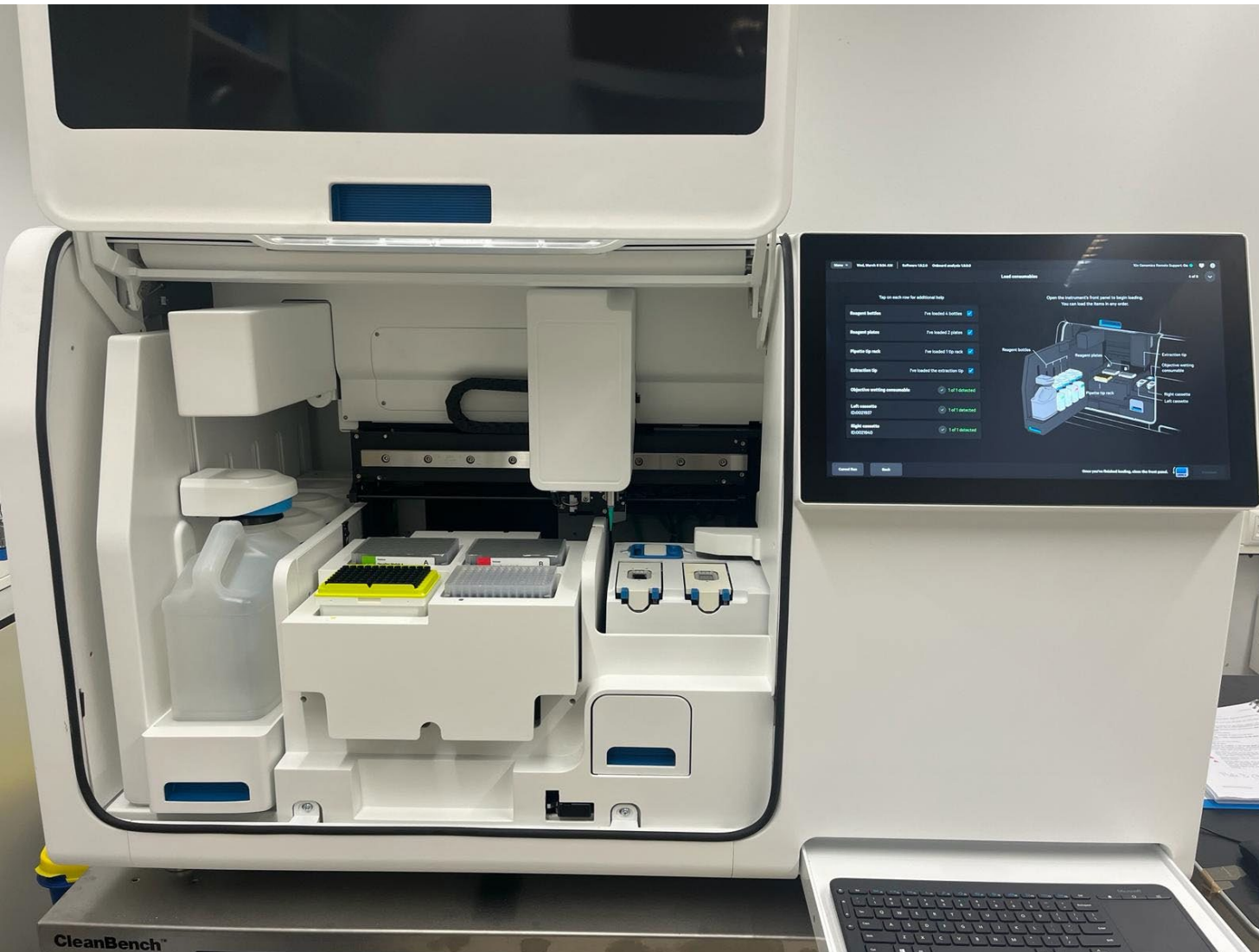
Spatial sequencing



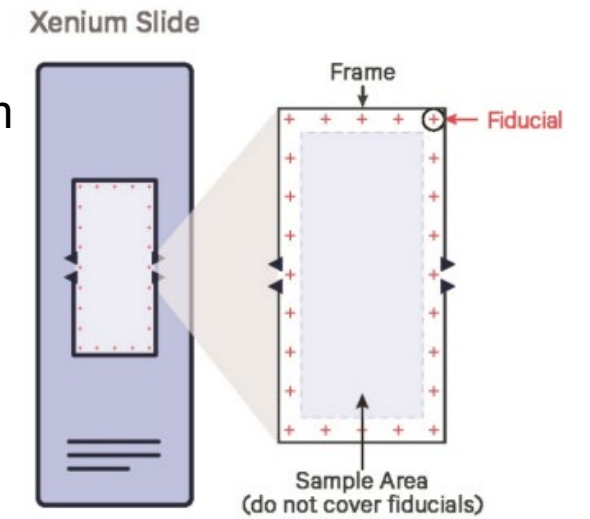
Spatial tagging



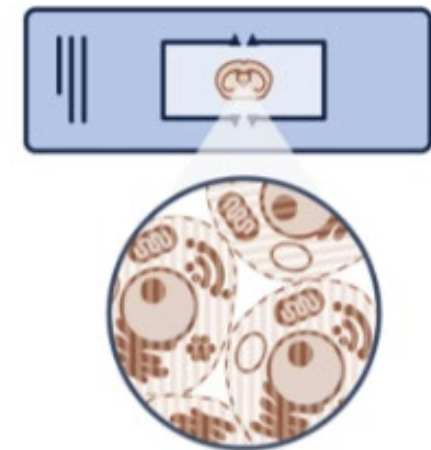
Xenium



Sample area:
10,45 X 22,45 mm

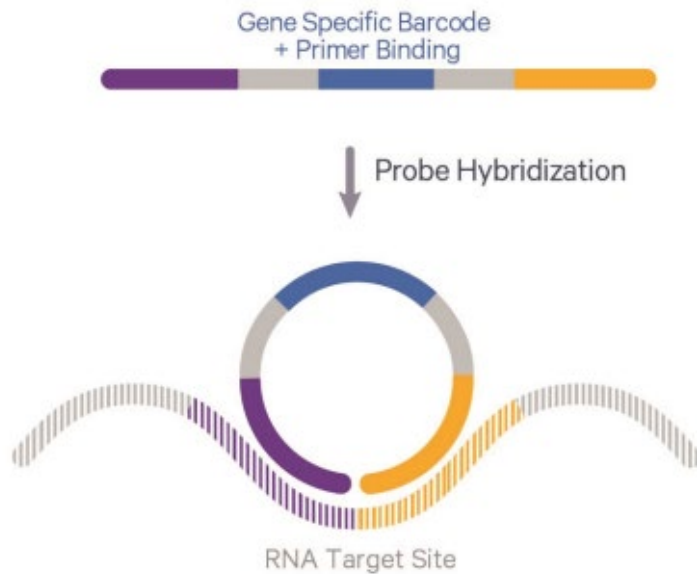


**Fixation & Permeabilization (FF) or
Deparaffinization & Decrosslinking (FFPE)**

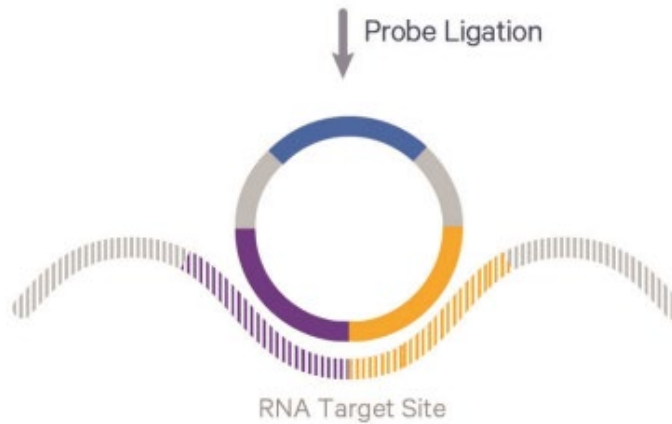


Xenium – how it works

Padlock probe hybridization



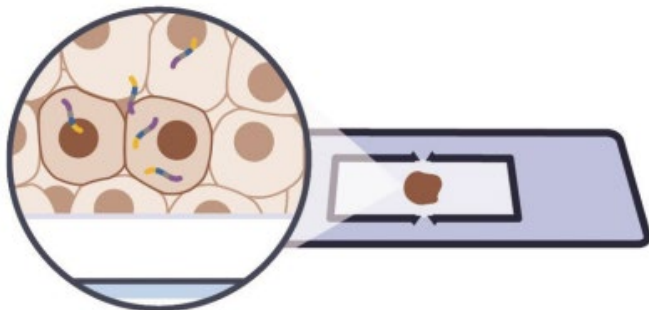
Probe ligation



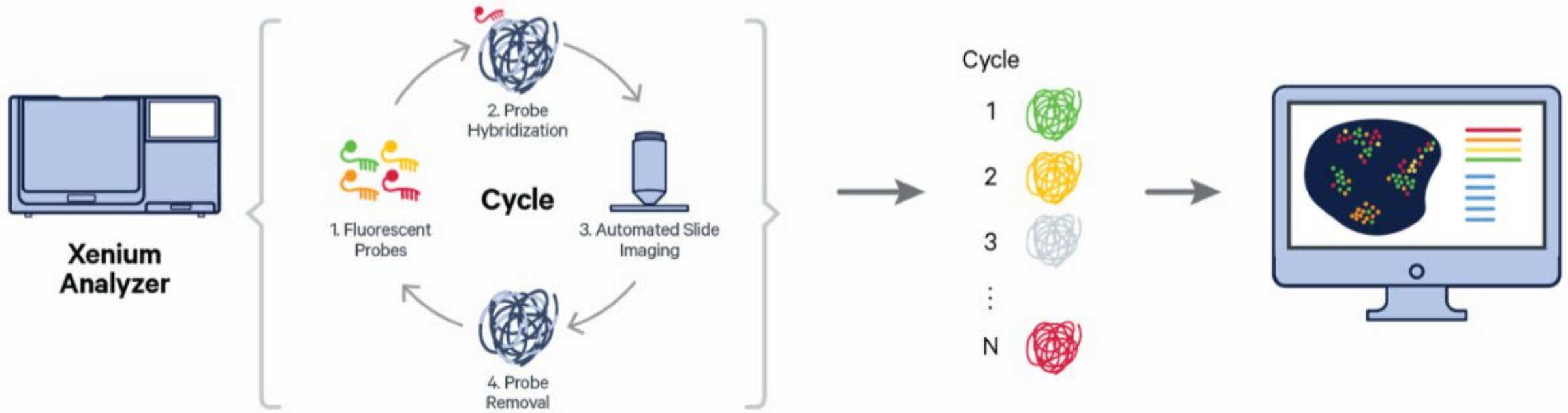
Probe amplification



- Left and right gene recognition region
- Gene specific barcode (with binding regions for fluorescent-labeled oligos)



Fluorescent probe hyb, imaging and decoding



Sequencing *in situ*



Methods: in situ sequencing (ISS), STARmap
Commercial: Xenium, StellarOmics, Singular G4x



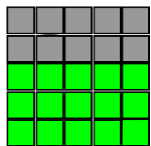
Resolution: Amplicon size (0.5 - 1µm)



Detection efficiency: 10 - ~50%



Gene throughput: 10 - 1,000

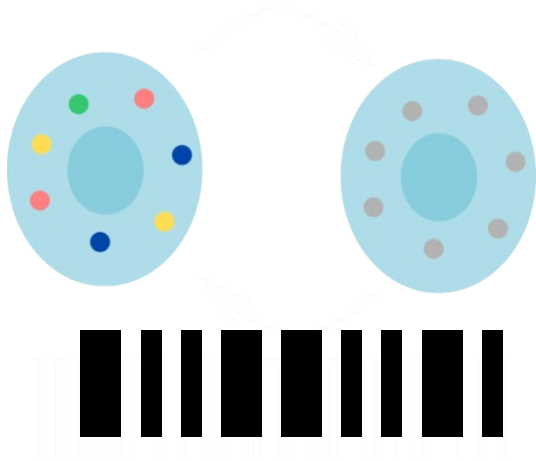


Spatial throughput: several mm² - several cm²

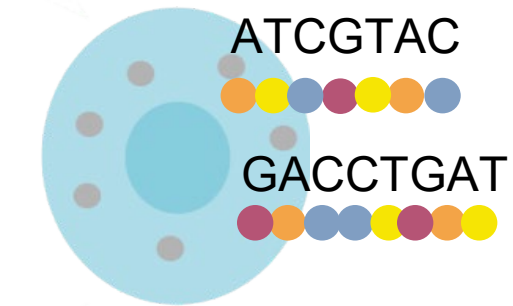
Spatial RNA Detection

Microscopy

Barcoded FISH

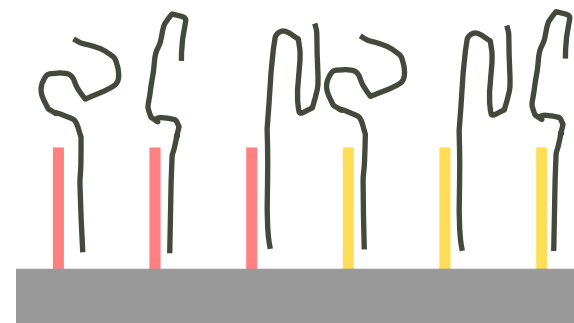


in situ sequencing

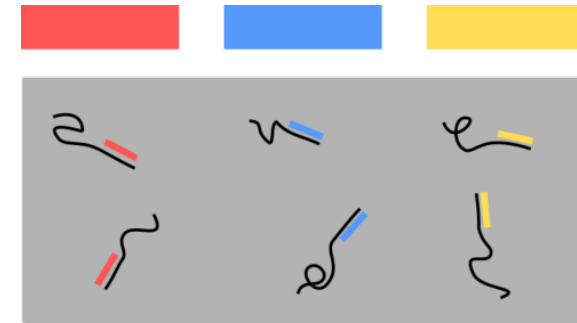


Sequencing

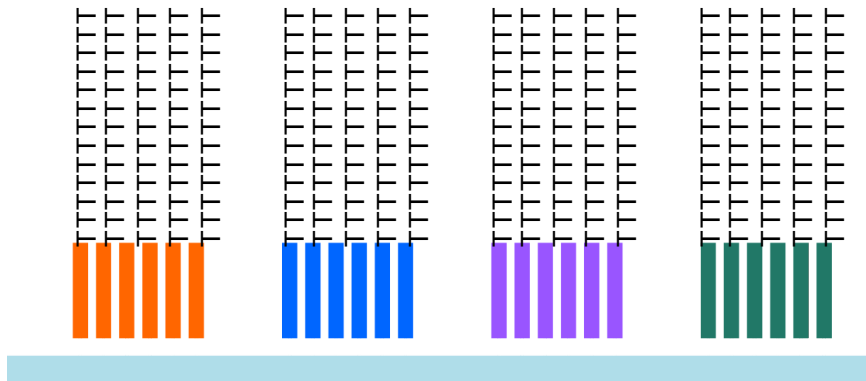
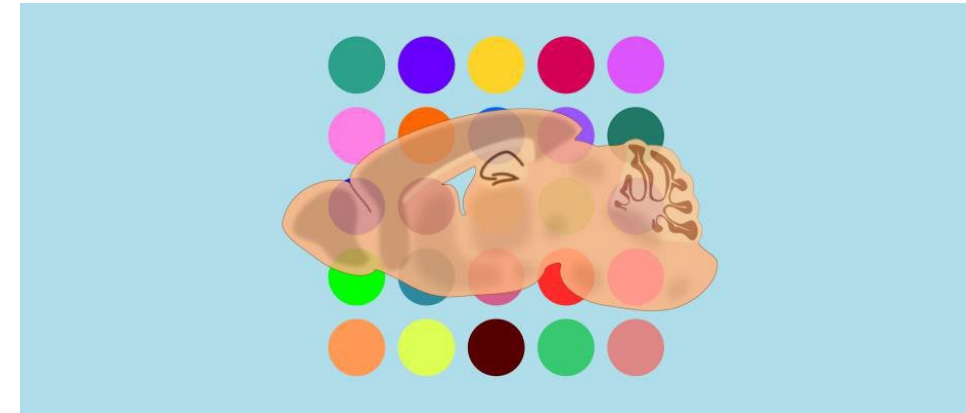
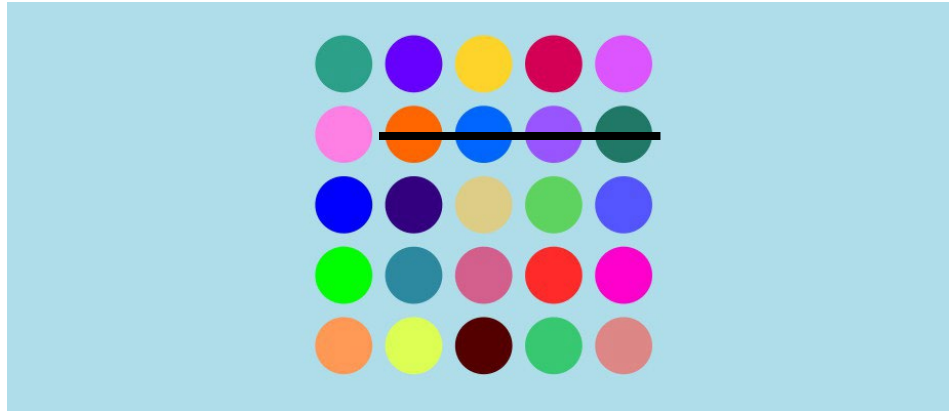
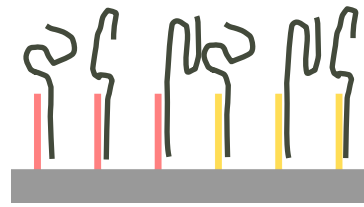
Spatial sequencing



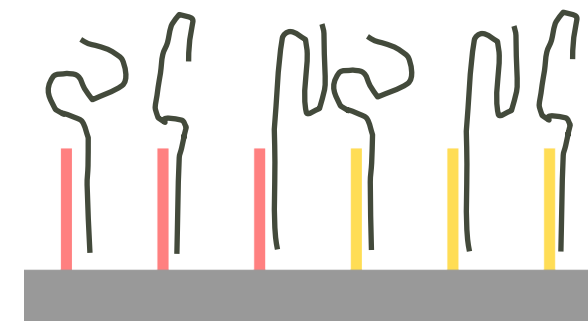
Spatial tagging



Spatial transcriptomics



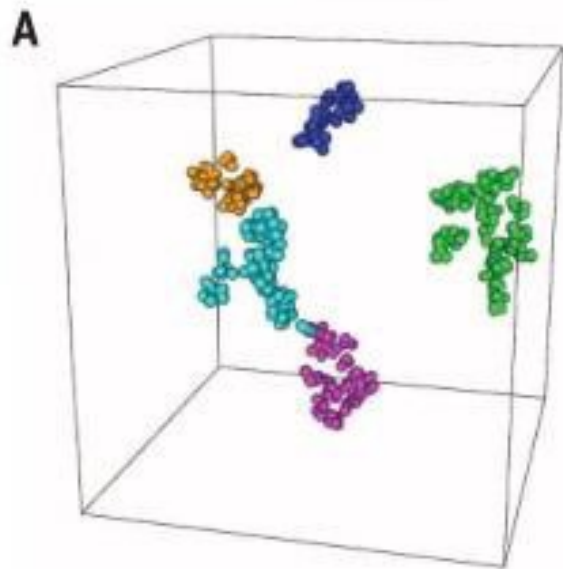
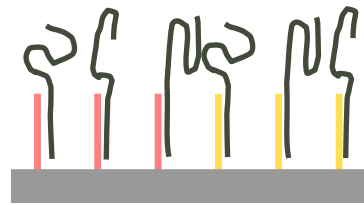
Microarray with spatial barcodes



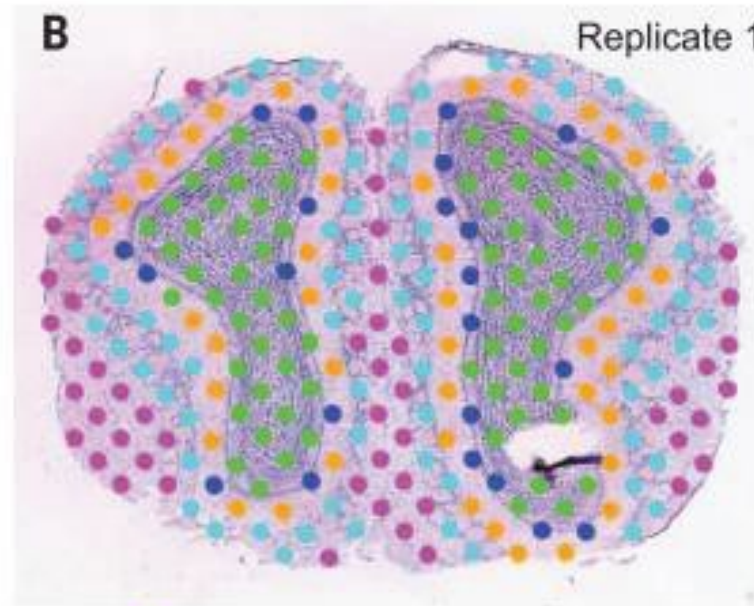
Stahl *et al.* Science 2016



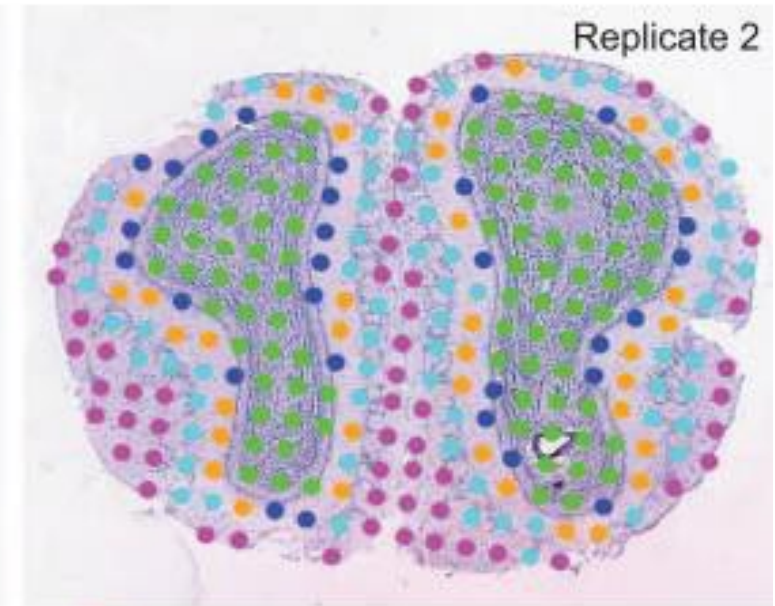
Spatial transcriptomics



- Cluster 1: Granular cell layer
- Cluster 2: Mitral cell layer



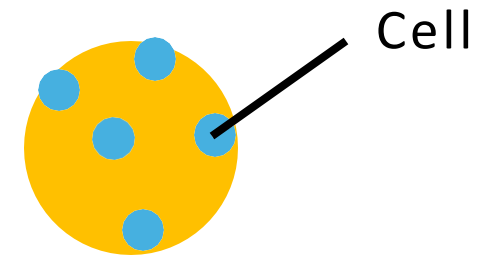
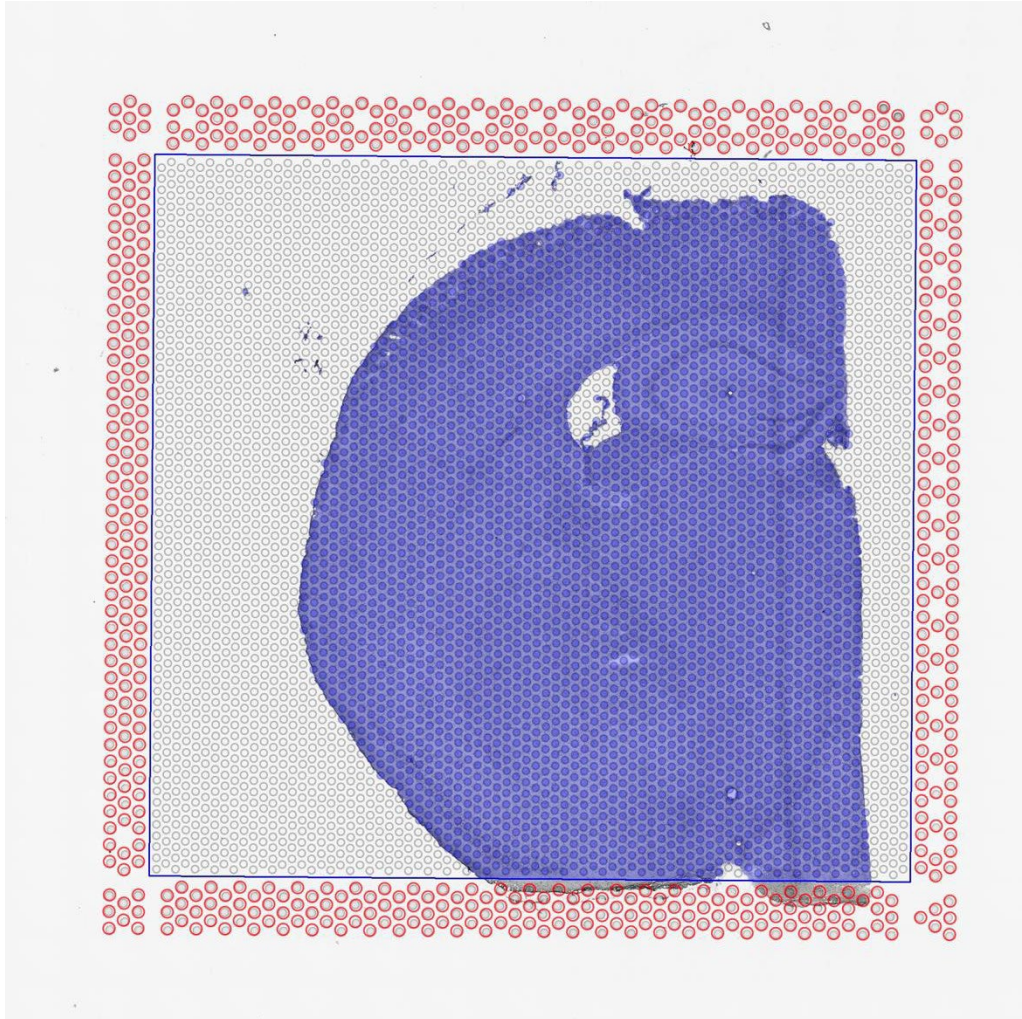
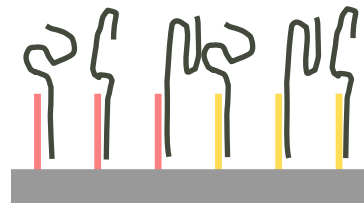
- Cluster 3: Outer plexiform layer
- Cluster 4: Glomerular layer



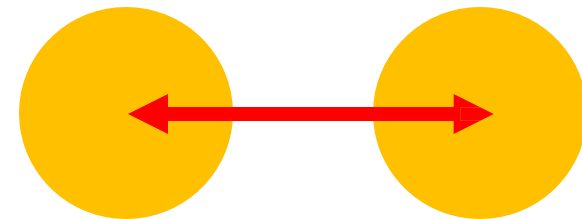
- Cluster 5: Olfactory nerve layer

Stahl *et al.* Science 2016

10X Genomics Visium

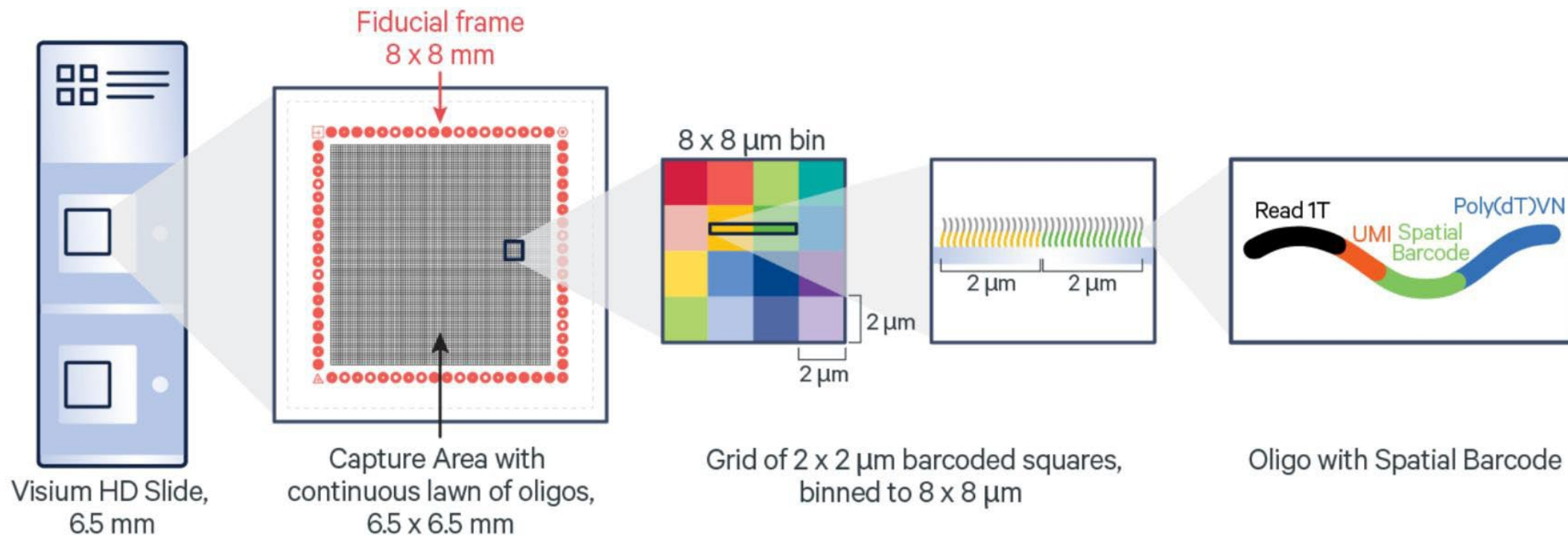


Spot size: 55 μm



Spot spacing: 100 μm

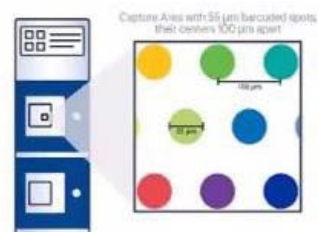
10X Genomics Visium HD



10X Genomics Visium HD



Visium v2

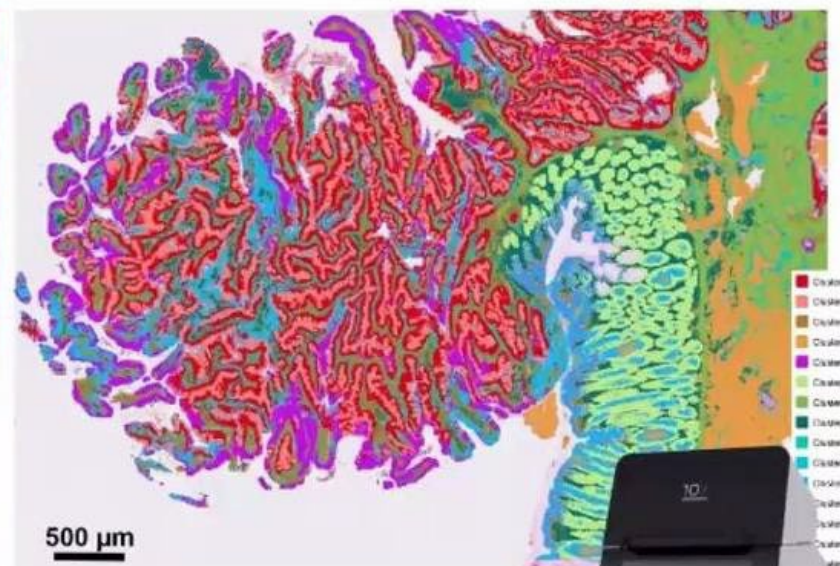


5000 spots
55µm resolution
spaced 100µm



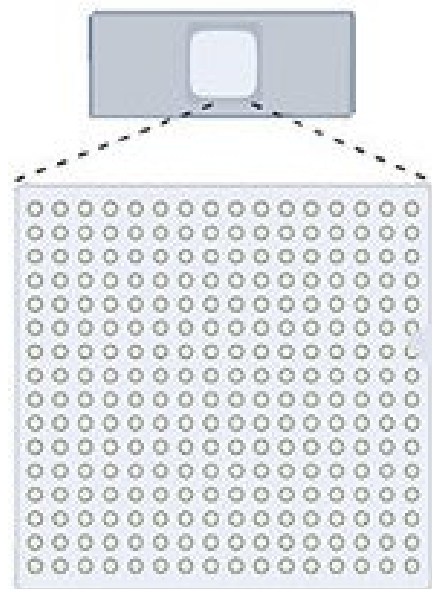
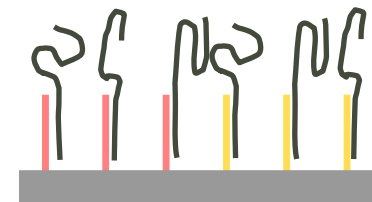
Visium HD

11M squares
2x2 µm resolution
Continuous coverage

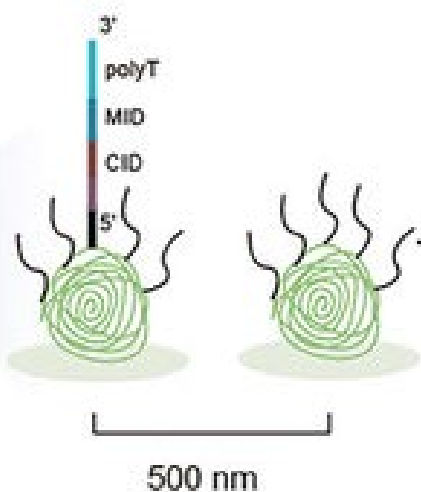


10X Visium HD

Stereo-seq

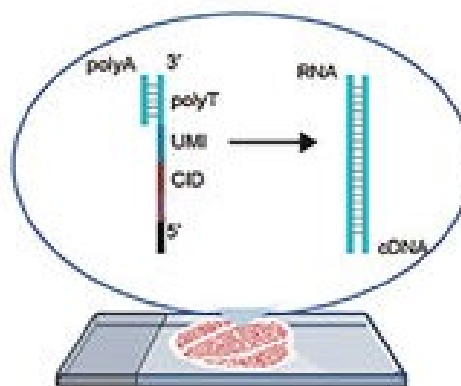


Chip

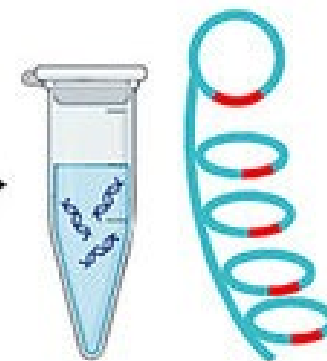


Capturing probe

220nm DNA nanoballs



Reverse Transcription

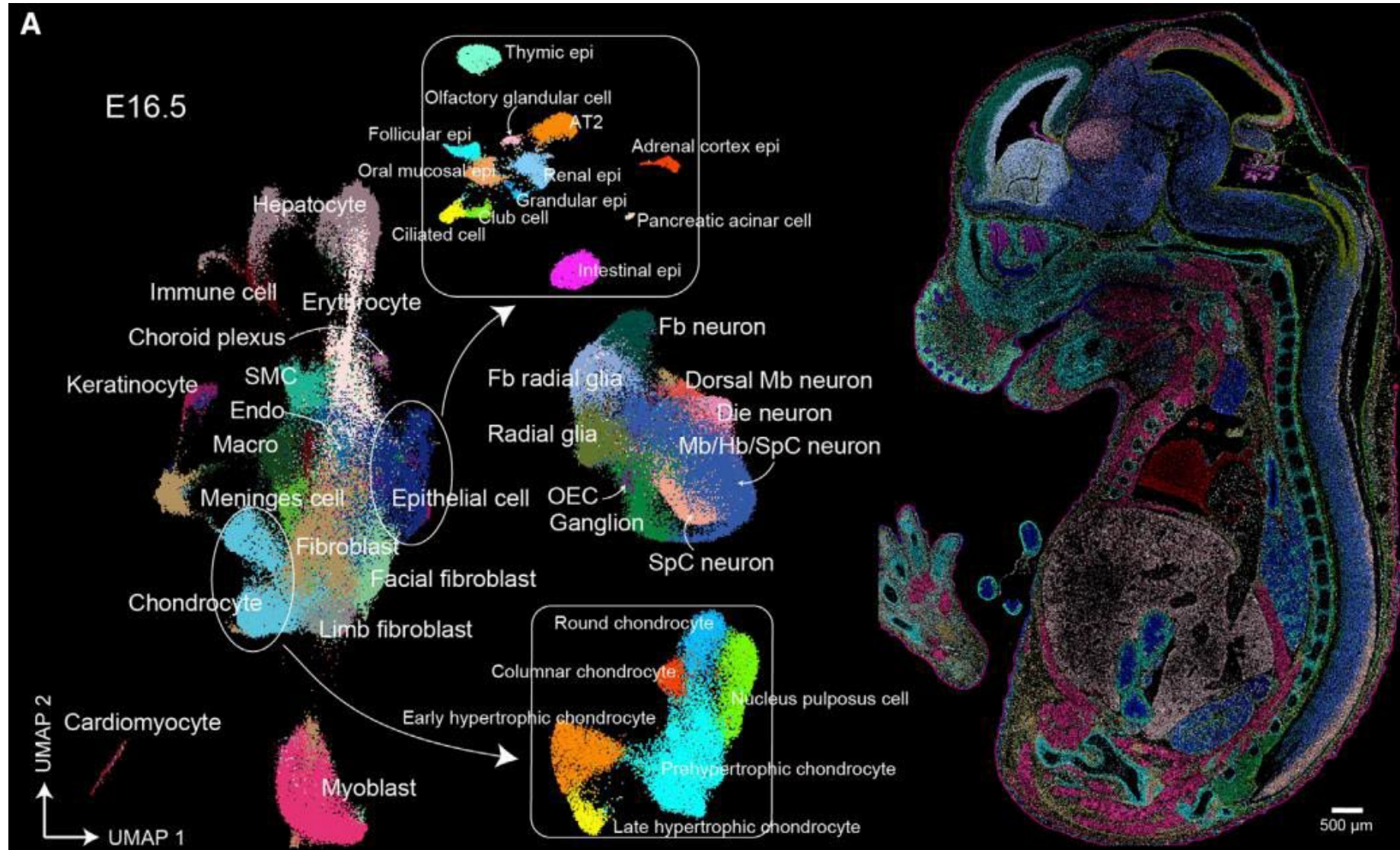
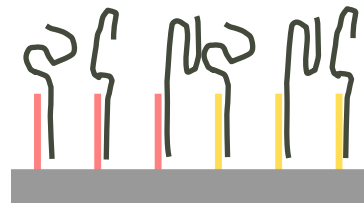


Library construction

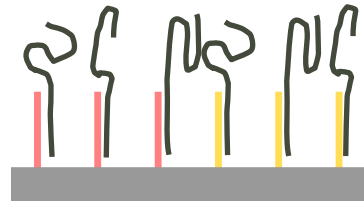


Sequencing

Stereo-seq



Spatial Sequencing



Methods: Spatial Transcriptomics, Slide-seq

Commercial: 10X Visium, STOmics Stereo-seq, Takara/Curio Seeker



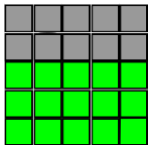
Resolution: Spot size 220nm - 100um (but RNA diffuses)



Detection efficiency: 0.1 - 5%



Gene throughput: Full transcriptome

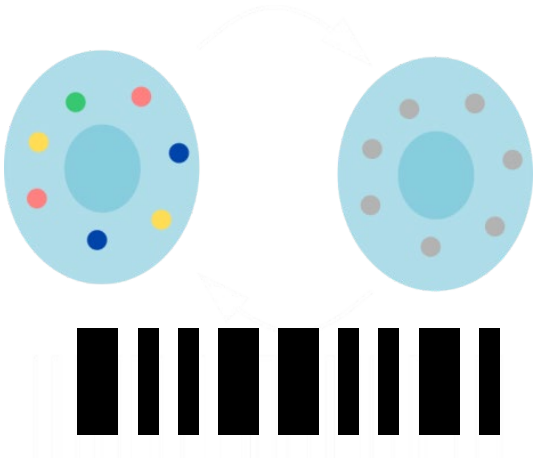


Spatial throughput: several mm² – cm²

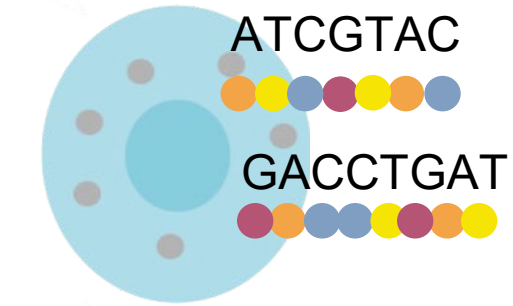
Spatial RNA Detection

Microscopy

Barcoded FISH

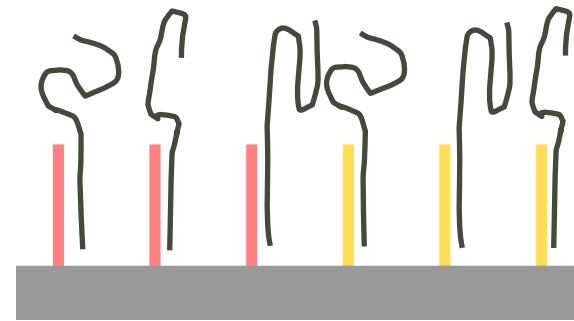


in situ sequencing

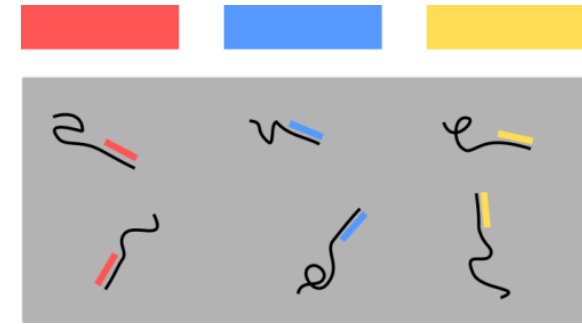


Sequencing

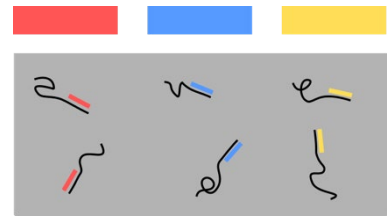
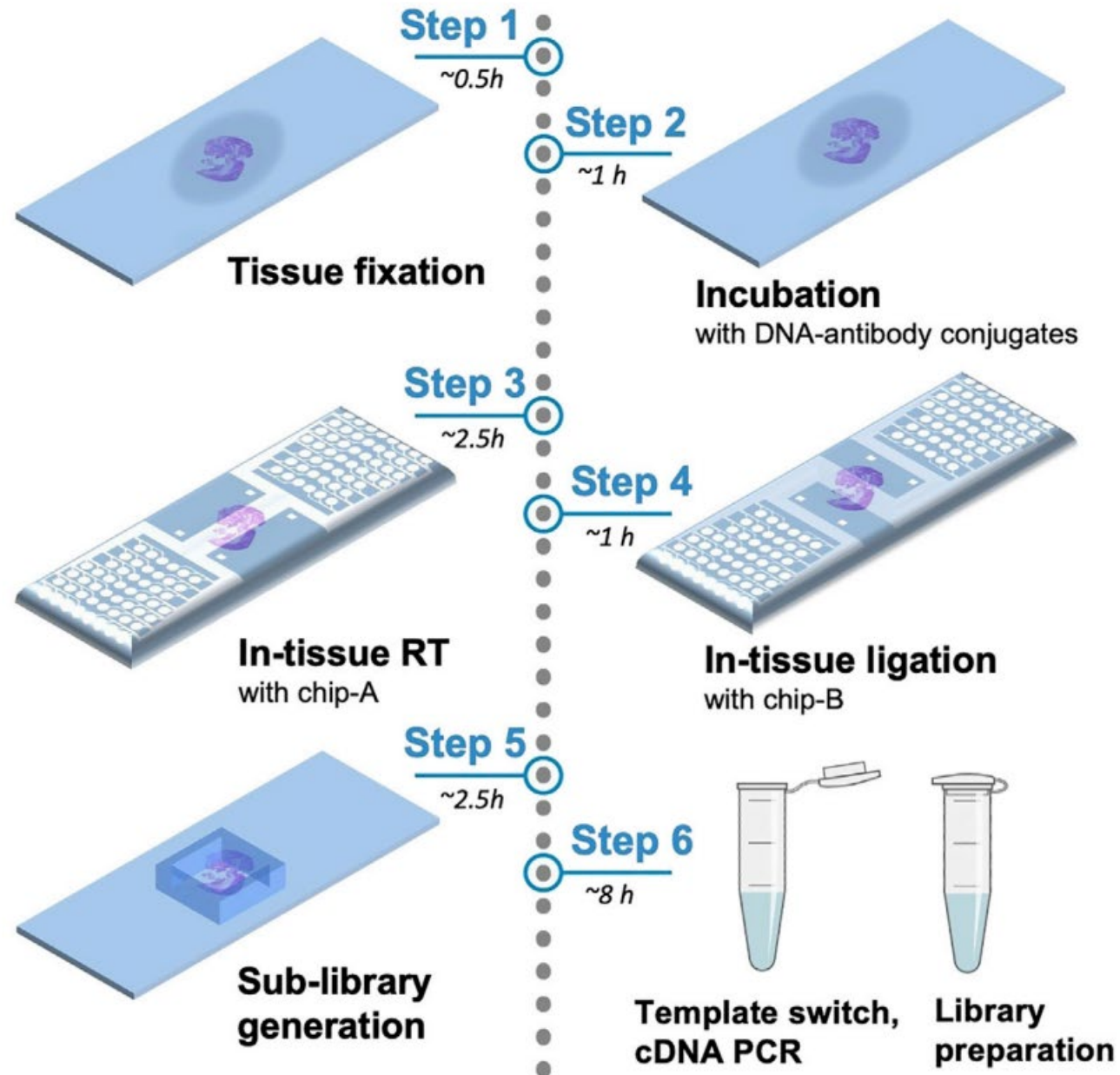
Spatial sequencing



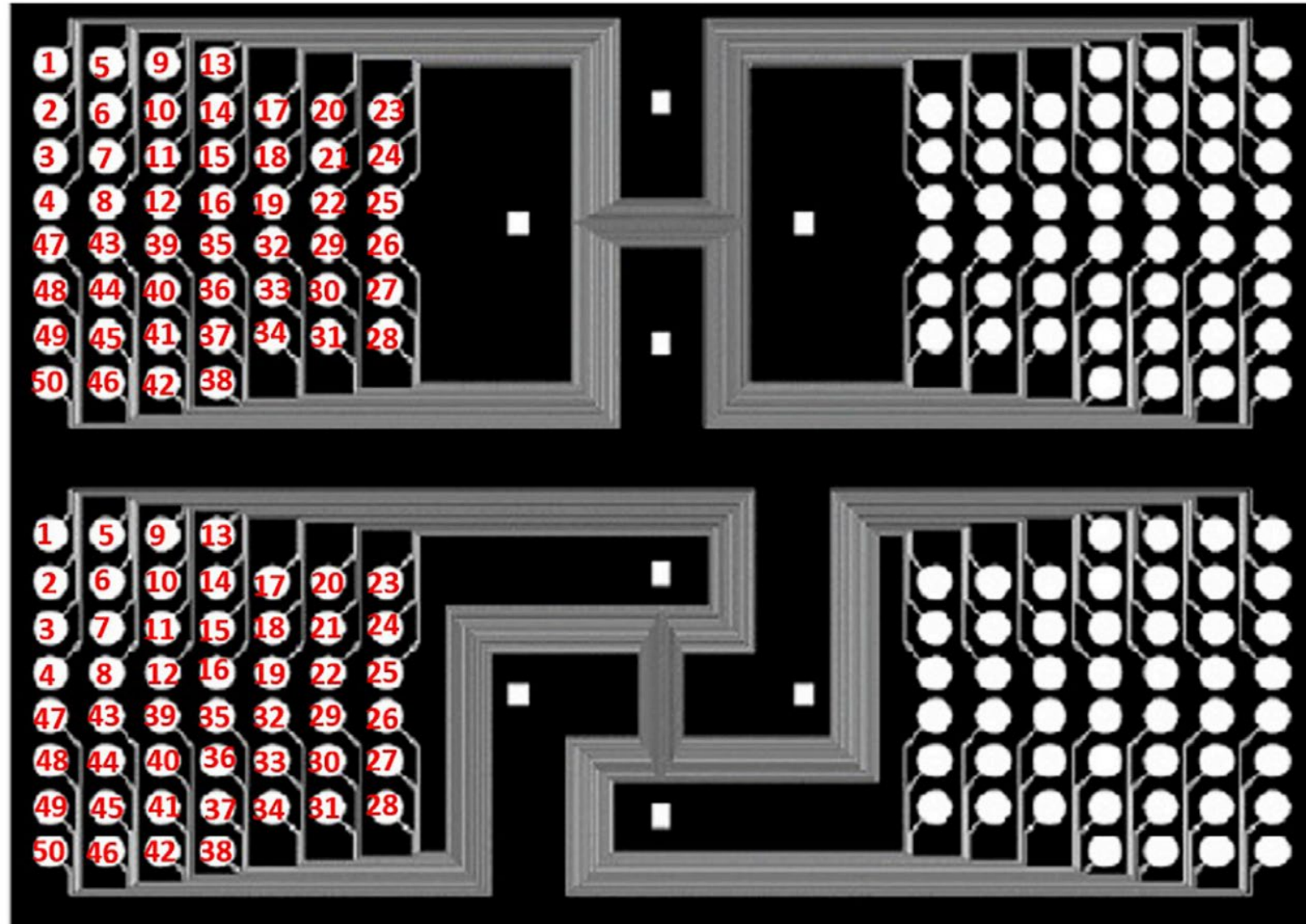
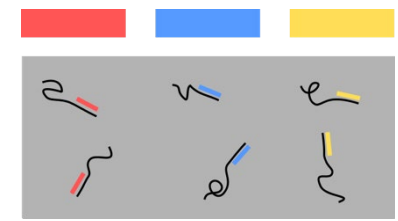
Spatial tagging



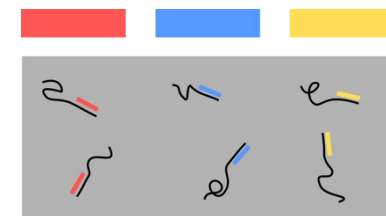
DBiT-seq



DBiT-seq

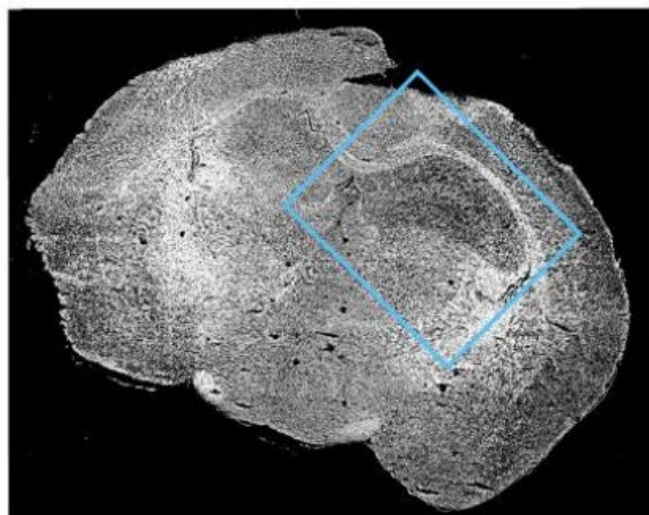


AtlasXomics



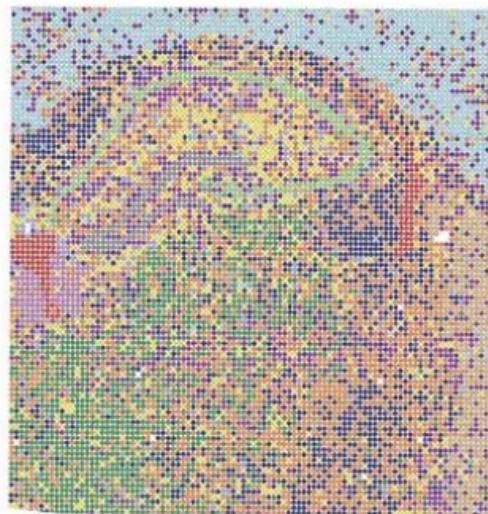
Our platform : Unbiased spatial epigenome mapping at cellular level

Histology Image



Adult mouse coronal section

Open Chromatin



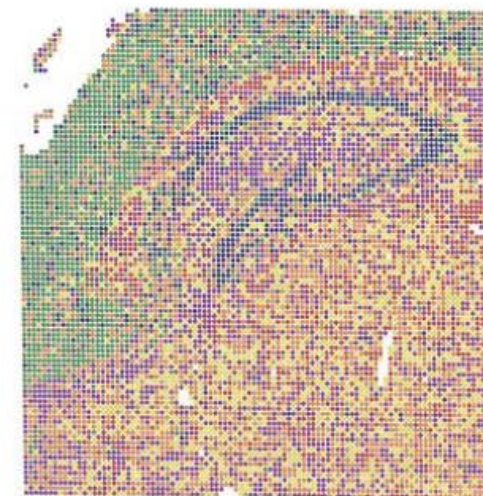
Spatial ATAC-seq

Histone Modifications



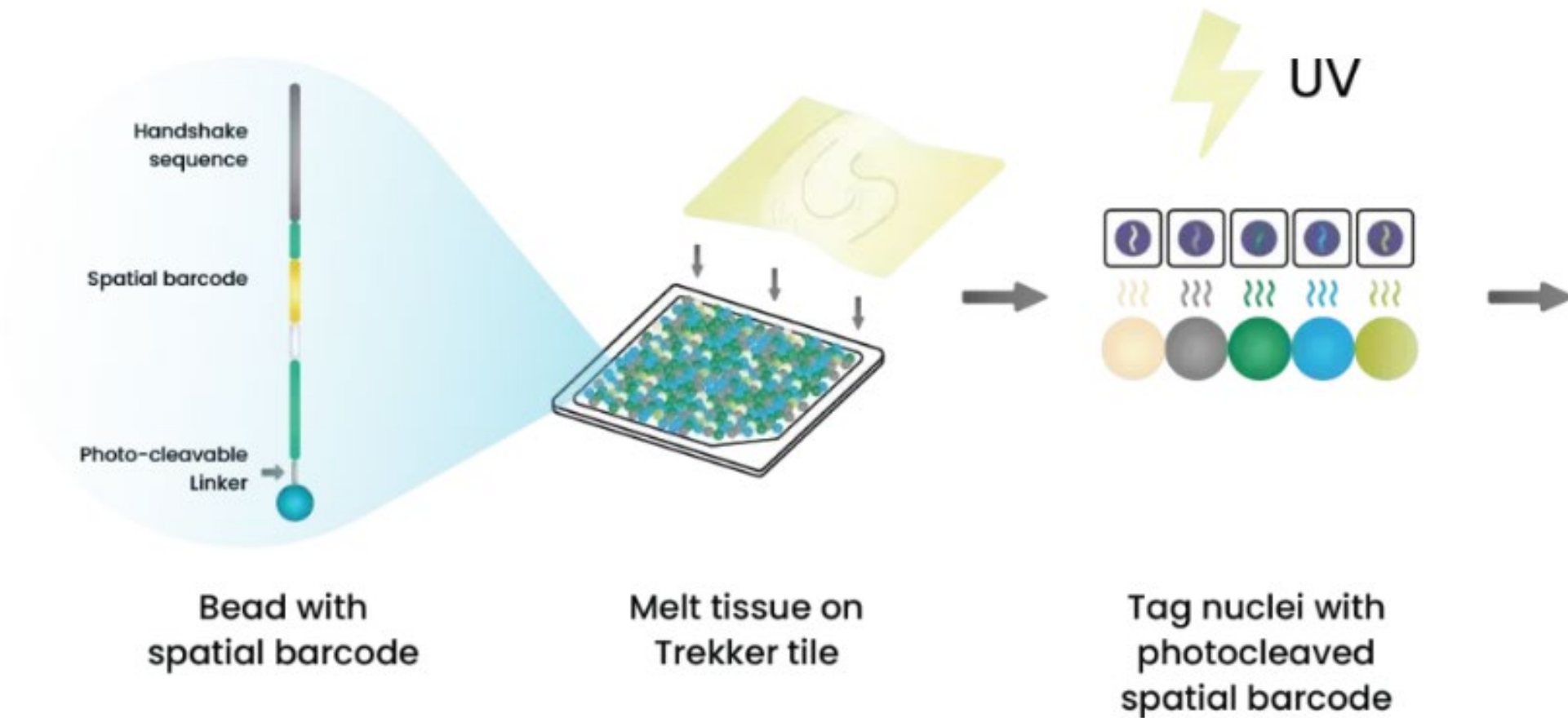
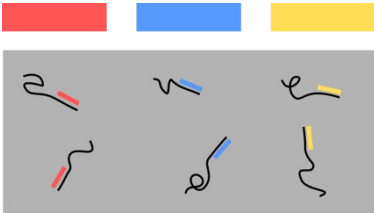
Spatial H3K27ac

Chromatin Associated Proteins

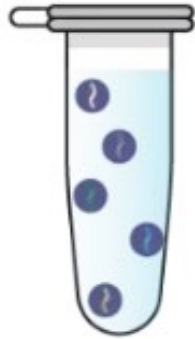
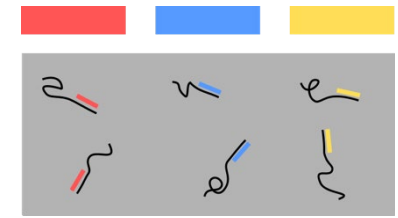


Spatial RNA pol II

Slide-tags



Slide-tags



Isolate nuclei

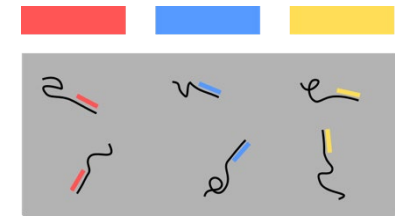


Single nuclei RNA-seq



Trekker bioinformatics analysis
places nucleus in space based
on spatial barcodes associated
with nucleus

Spatial Tagging



Methods: DBiT-seq, Slide-tags

Commercial: AtlasXomics, Takara/Curio Trekker



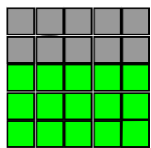
Resolution: 10 – 100 μ m



Detection efficiency: 1 - 30%



Gene throughput: Full transcriptome

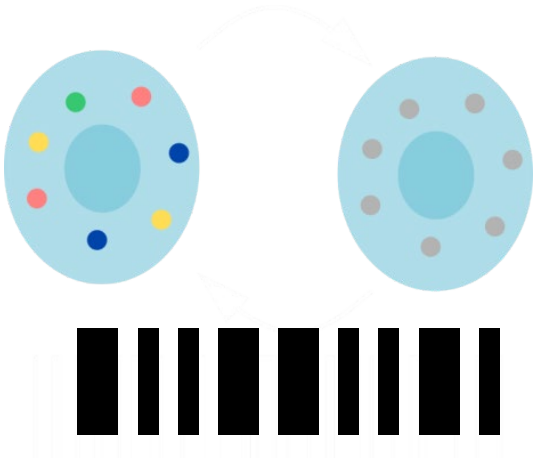


Spatial throughput: several mm² – cm²

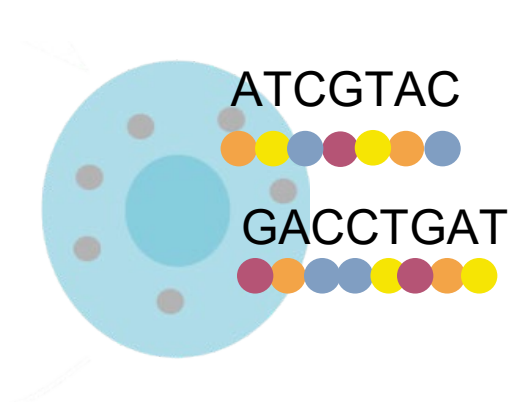
Spatial RNA Detection

Microscopy

Barcoded FISH

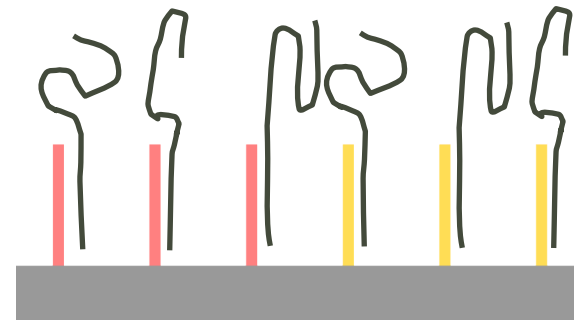


in situ sequencing

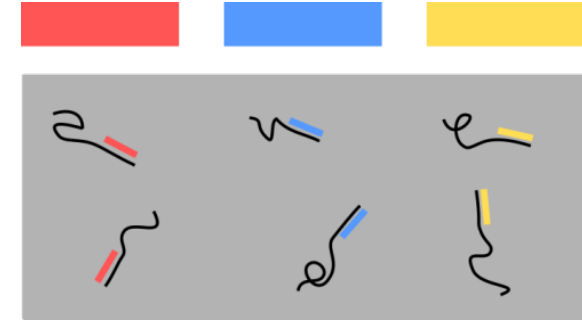


Sequencing

Spatial sequencing



Spatial tagging



Further reading

Review | [Open access](#) | Published: 20 January 2025

A practical guide for choosing an optimal spatial transcriptomics technology from seven major commercially available options

[Hyun Ju Lim](#), [Ye Wang](#), [Anton Buzdin](#) & [Xinmin Li](#) 

[BMC Genomics](#) **26**, Article number: 47 (2025) | [Cite this article](#)

Questions?

